

Project Proposal

Systematic Evaluation of LLM Planning Capabilities Across Multiple Domains and Model Families

Course: **AI in Industry** — Artificial Intelligence, Università di Bologna
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1. Context

Two prior student projects investigated whether LLMs can solve PDDL planning problems using iterative validator-guided feedback. Project A (Merola et al.) evaluated LLaMA 3.3 70B across many domains (Blocks World, Gripper, Logistics, Hanoi, etc.) and built meta-networks for confidence estimation. Project B (D’Ascenzo & Gentili) compared four models (LLaMA3-8B, Phi4-14B, Gemma3-27B, Kimi-Dev-72B) on two domains (City Car, Tetris) with a modular HPC-ready framework.

Both share the same core architecture, LLM planner + VAL validator + feedback loop, but neither provides a unified cross-model, cross-domain picture. Our project merges their work into a single coherent experimental study.

2. Objectives

We have following objectives:

- Unify the two codebases into a single repository with a consistent pipeline for model loading, plan generation, validation and results reporting.
- Run a controlled comparative study of 3 LLM families across 3 planning domains, using standardized prompts and iterative refinement (up to 4 rounds).
- Address the prompt-bias confound by using a single unified prompt across all models so the performance differences reflect model capability rather than prompt tuning.
- Produce a clear comparative analysis with success rates, iteration convergence patterns and domain-specific insights that extend the findings of both prior projects.

3. Experimental Design

3.1. Domains (3 minimum)

We select three domains that represent distinct reasoning challenges, drawn from the existing problem sets:

Domain	Reasoning Challenge	Source
Blocks World	Sequential state-space search, constraint chaining	Project A
City Car	Multi-agent coordination, infrastructure mgmt.	Project B
Tetris	Spatial/geometric reasoning	Project B

3.2. Models (3 families)

We retain one model from each prior project for continuity and add one newer model for relevance. Our candidate set, subject to compute availability:

Model	Rationale	Origin
LLaMA 3.3 70B	Strong baseline from Project A	Retained
LLaMA3-8B	Tests scale effect within same family	Retained from Project B
Qwen 2.5 / Mistral / DeepSeek	Different arch. family, recent release	New addition

3.3. Prompting & Refinement

To resolve the prompt-bias issue from Project B, we use a single unified prompt (system + problem template) across all models. We test two prompting conditions:

- Baseline: Direct PDDL problem description with structured output instructions (zero-shot).
- Chain-of-Thought: Same prompt with a reasoning-step prefix encouraging step-by-step plan construction.

Both conditions include iterative refinement: up to 4 rounds of VAL validator feedback, with all intermediate plans and validation logs recorded.

3.4. Evaluation Metrics

For each model–domain–prompt combination:

- Success rate (% of problems solved within 4 iterations)
- Average iterations to convergence (for solved problems)
- Iteration gain curve (% of solutions at iteration 1, 2, 3, 4)
- Per-problem features: valid action %, consecutive valid steps, logical violations, plan length (for dataset construction)

4. Questions for the Supervisor

- 4.1. Compute access: Is CINECA/Leonardo available to our group? If not, we will scope down to models runnable on consumer GPUs (8B–14B range).
- 4.2. Data reuse: Can we incorporate raw results from the prior projects into our comparative analysis, or should all experiments be re-run from scratch?
- 4.3. Deliverable format: Is the expected output a report + repository, an oral presentation, or both?